Fiscal Year:	FY 2017	**Task Last Updated:**	FY 04/01/2017
PI Name:	Mueller, Holger Ph.D.		
Project Title:	QWEP Analysis and Optimization		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-0001	**Congressional District:**	9
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Extended to 06/28/2018 per U. Israelsson/JPL (Ed., 4/3/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kohel, James (Jet Propulsion Laboratory) Yu, Nan (Jet Propulsion Laboratory)		
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Task Description:

The equivalence of gravity and accelerated motion is the basis of general relativity's description of gravity as a result of curved space and time. At the same time, dark matter and dark energy show that our understanding of physics is incomplete, and that new discoveries are needed to explain the dominant constituents of the universe. Perhaps these are hints that the standard description of gravity must be modified.
This description, however, is incompatible with the quantum theories of the electroweak and the strong interactions, in which space and time exist independently of the forces. It is therefore not surprising that unified theories of gravity and quantum mechanics are expected to contain tiny violations of the gravity-acceleration equivalence. Such violations would manifest themselves as variations between the acceleration of free fall for different types of matter. These variations are considered as being among the most promising areas to look for signals arising from a unification of general relativity and quantum mechanics. ESA's (European Space Agency) quantum weak equivalence principle (QWEP) mission seeks such signals with a strongly improved sensitivity of 10-15 or better. By using quantum interference of ultracold atoms, it will use quantum particles where previous tests used bulk matter.

The research proposed here will combine theory and experiment to maximize the sensitivity and scientific impact of QWEP, and minimize the risk. Using the standard model extension (SME) as a theoretical framework, we will study which species of atoms, what quantum states, and what type of analysis might lead to the largest and most interesting signals. The choice of species, however, is equally dependent on experimental considerations. Using our existing atom interferometers, we will experimentally compare different technologies for the matter-wave beam splitters that are suitable for the candidate species, and develop strategies to minimize systematic effects such as rotation and gravity gradients. Based on the experience of Jet Propulsion Laboratory (JPL) and the Space Sciences Laboratory (SSL) at University of California (UC) Berkeley, we will match these technologies to the requirement of space operation. Thereby, this work will apply fundamental theory, atomic physics experiment, and principles of astronautics to a comprehensive study optimization and analysis of the QWEP mission.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

We seek to understand two things: What is the difference between a quantum and classical test of the weak equivalence principle? What kind of quantum test would provide the most new information regarding Einstein’s principle of equivalence? We have made progress on answering these questions, which have been pursued both theoretically and experimentally.
We have also started the effort to look for hints of dark energy in atom interferometry experiments. These experiments have started an international effort looking for new models of dark matter and dark energy, and promise to either discover well-motivated theories, or to rule them out once and for all.

Task Progress:

Our work in the past year had three main thrusts. In the first thrust, we have been investigating how spaceborne atom interferometry can probe models for dark matter and dark energy. This has resulted in experimental demonstrations of such tests for so-called chameleon and symmetron models [1], and theoretical studies on how to make detailed predictions of the putative signals [2]. In the second thrust, we have developed strategies to overcome systematic effects in atom interferometers that use Bragg diffraction, as the one planned for the Cold Atom lab [3, 4]. In the third, we have developed specific plans for spaceborne atom interferometry. We have collaborated with Nick Bigelow and the German team and written the Science Requirements Document for atom interferometry on the Cold Atom Lab. Plans include demonstrating long coherence times thanks to microgravity, tests of the equivalence principle, and searches for dark-energy candidates. We have also collaborated with Nan Yu (JPL) on a concept study QTEST (Quantum test of the Equivalence Principle and Spacer-Time) for testing the equivalence principle in space [5].
After running our laboratory experiment over 170 hours (3 weekends), we were able to resolve the gravitational pull from a 0.19 kg source mass on the atoms within an uncertainty of more than three sigma, which is equal to a force of ~10^-32 Newton. This measurement puts strong limits on screened fifth forces. A publication which explains how these data constrains on screened scalar field theories, which are intended to explain the accelerated expansion of the universe, is in preparation.

We also started studying blackbody radiation as an effect in atom interferometry. Space-based atom interferometers may be subject to large environmental temperature gradients, especially if atoms outside a spacecraft are used, which would be exposed to thermal radiation from celestial bodies. These give rise to gradients in the intensity of blackbody radiation. Through the AC Stark shift, blackbody radiation gradients will induce accelerations to the atoms that could be as large as several nanometers per second squared. We used our cavity-based atom interferometer to study this effect; the central mass was heated with an infrared laser and an acceleration of the atoms towards the body was observed.

We observed a temperature-dependent attractive force on cesium atoms towards a cylinder heated to up to 100C in a vacuum chamber which is otherwise at room temperature. We find the force proportional to the fourth power of the source temperature, as is characteristic of thermal radiation, and explain it by the potential caused by the AC Stark shift to the ground state of atoms bathing in blackbody radiation. While the thermal black body radiation field possesses a purely outward directed Poynting vector the attractive dipole force in the experiment largely dominates radiation pressure for temperatures up to a few thousand Kelvin. We find the force to be in agreement with theoretical predictions. It can be much stronger than gravity and will affect both high precision tests of the equivalence principle and adds a new optical force component to the radiation pressure on atoms in astrophysical scenarios.

[1] Atom-interferometry constraints on dark energy. Paul Hamilton, Matt Jaffe, Philipp Haslinger, Quinn Simmons, Holger Müller, and Justin Khoury, Science, 349, 849 (2015)

[2] Chameleon Dark Energy and Atom Interferometry. Benjamin Elder, Justin Khoury, Philipp Haslinger, Matt Jaffe, Holger Müller, Paul Hamilton, PRD 94, 044051

[3] Controlling the Multiport Nature of Bragg Diffraction in Atom Interferometry. Richard H. Parker, Chenghui Yu, Brian Estey, Weicheng Zhong, Eric Huang, and Holger Müller, Phys. Rev. A 94, 053618 (2016)

[4] High resolution atom interferometers with suppressed diffraction phases. Brian Estey, Chenghui Yu, Holger Müller, Pei-Chen Kuan, and Shau-Yu Lan, Phys. Rev. Lett. 115, 083002 (2015)

[5] Quantum Test of the Equivalence Principle and Space-Time aboard the International Space Station. Jason Williams, Sheng-wey Chiow, Holger Müller, and Nan Yu, New J. Phys, 18 (2016)

Bibliography Type:	Description: (Last Updated: 01/29/2019)
Articles in Peer-reviewed Journals	Elder B, Khoury J, Haslinger P, Jaffe M, Mueller H, Hamilton P. "Chameleon dark energy and atom interferometry." Physical Review D. 2016 Aug 15;94(4):044051. https://doi.org/10.1103/PhysRevD.94.044051 , Aug-2016
Articles in Peer-reviewed Journals	Parker RH, Yu C, Estey B, Zhong W, Huang E, Mueller H. "Controlling the multiport nature of Bragg diffraction in atom interferometry." Physical Review A. 2016 Nov; 94(5):053618. https://doi.org/10.1103/PhysRevA.94.053618 , Nov-2016